THE PATH TO CARBON ZERO

India is well on its way to meet its target of net zero emissions by 2070. Meanwhile, it needs to tap its natural resources and leverage the renewable power generation infrastructure to become a world leader in the supply of renewable energy

By **Jumana Shah**

When it comes to going green, India may well be making others go green—with envy. The country may be the third-largest carbon emitter in the world in terms of absolute numbers, but it also has the lowest per capita GHG (greenhouse gas) emissions. Now, it is well on its way to achieve its committed target of ensuring that 50 per cent of the country's energy is met from decarbonised sources and there is more than 500 GW of green power generation by 2030. Not only is this remarkable in itself but India seems to be doing much better than other developed countries like the UK and Germany, for instance, in meeting its Nationally Determined Contributions or NDCs. The two countries have, in fact, dialled down their targets in the wake of the pressure on their economies. India plans to hit net zero emissions by 2070, the target set at the most recent global climate change meet.

It is from here on, however, that India's journey will get tricky. For one, our energy needs are expected to grow twofold by 2047. The country's peak power demand is estimated to touch 400 GW in 2031-32 from 245.2 GW currently. Green energy is also expensive, though India is banking on technology to make its production cheaper. Then, there is the timeline. As Czech-Canadian scientist and policy analyst Vaclav Smil says, a "widespread transition from one dominant fuel to another typically takes 50-60 years. It will require generations of perseverance". Finally, for almost all the alternative energy sources available currently, India is dependent on import of raw materials critical to the process as also the machinery, even as China has become an indispensable source of critical minerals and rare earths.

"The answer thus lies in the *aatmanirbharta* of green resources," says Kartikeya Sarabhai, director of Center of Environment Education in Ahmedabad. We must also invest in indigenous research and development (R&D) while achieving economies of scale in manufacturing within the country to offset the additional cost of shifting to newer evolving technologies. These are on the lines of the government's chosen path too, based on its policy measures. "India has been deploying clean energy technologies at an accelerated pace since 2009," says Rishabh Jain, senior programme lead for Technology Futures programme at CEEW (Council on Energy, Environment and Water). "Thanks to India's natural resources and endowments, there is huge potential for deploying clean energy technologies such as solar, wind (onshore and offshore), hydro, bioenergy and green hydrogen. When deployed at scale, there can also be economic opportunities for manufacturing the key components domestically, creating new jobs and economic growth. China has been manufacturing clean energy technologies for the whole world, but countries are now looking

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